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Morphology-Aware Logit-Level Lexically Constrained Decoding for English–Turkish Neural Machine Translation

Alkim Gönenç Efe, and Emre Şatır

Department of Computer Engineering, Izmir Katip Celebi University, Izmir, Turkey
efe2004alkim@gmail.com, emre.satir@ikcu.edu.tr

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Abstract

Lexically constrained decoding allows users to inject domain-specific terminology into Neural Machine Translation (NMT) output at inference time, without retraining. While prevailing approaches modify the structural beam search algorithm—such as Grid Beam Search and Dynamic Beam Allocation (DBA)—they often introduce severe computational overhead, complex multi-bank state tracking, and heightened susceptibility to ungrammatical sequence inflation when applied to morphologically rich agglutinative languages. In this paper, we propose a lightweight, plug-and-play, beam-search-agnostic logit manipulation framework that directly steers the token probability distribution at each decoding step through standard `LogitsProcessor` interfaces. We implement and evaluate six decoding strategies: hard exclusion, hard inclusion with dynamic anchor scheduling, soft logit penalty, curriculum-anchored soft reward with multi-tier escalation, and two combined modes. We conduct an extensive bidirectional empirical evaluation on English↔Turkish (EN↔TR) translation using the Helsinki-NLP OPUS-MT transformer models across a curated bilingual corpus of 500 sentence pairs (comprising 250 EN→TR and 250 TR→EN pairs), categorized by linguistic domain and difficulty. Hyperparameters are systematically optimized per translation direction using Optuna with a multi-objective scoring function that explicitly penalizes output length inflation. Our empirical findings demonstrate that curriculum-anchored soft reward with escalation achieves the superior quality–satisfaction balance across both translation directions: 99.2%/98.8% constraint satisfaction with unconstrained baseline BLEU preservation of 72.32/81.65 and near-natural length ratios of 1.062/1.058. Hard inclusion achieves 99.6%/99.2% satisfaction while executing 2.1–3.3× faster than the HuggingFace Dynamic Beam Allocation (DBA) baseline. Crucially, we identify a severe failure mode in structural beam allocation: DBA exhibits massive length inflation in EN→TR (length ratio 2.768), caused by finite-state beam traps arising from Turkish agglutinative morphology, whereas TR→EN exhibits milder inflation (1.289). We further characterize the *logit squeeze effect* in simultaneous constraint modes, parameter invariance in soft penalties via the *threshold suppression phenomenon* and provide token-level interpretability analyses of probability adjustments across generation steps.

Keywords: neural machine translation; lexically constrained decoding; logit manipulation; beam search; Turkish NLP; agglutinative morphology; hyperparameter optimization

1. Introduction

Transformer-based neural machine translation (NMT) architectures (Vaswani et al. 2017; Bahdanau et al. 2015; Sutskever et al. 2014) have established state-of-the-art benchmarks in cross-lingual fluency and general-domain translation quality. Nevertheless, deploying NMT systems in specialized professional environments—such as legal contracting, biomedical documentation, patent filing, technical manufacturing, and localization—imposes strict lexical requirements. In such domains, generic synonymy cannot be tolerated: a pharmaceutical translation must emit

precise, approved chemical nomenclature rather than colloquial variants; legal agreements necessitate rigid adherence to jurisdiction-specific terminology; and technical documentation demands absolute consistency across extensive product catalogs.

Enforcing vocabulary constraints dynamically at test time remains a foundational challenge in sequence-to-sequence modeling. Fine-tuning an existing model or retraining it from scratch on specialized corpora is computationally prohibitive for rapidly evolving customer glossaries and introduces the substantial risk of catastrophic forgetting regarding general syntactic competence. Consequently, inference-time *lexically constrained decoding* has emerged as an indispensable paradigm, seeking to enforce positive inclusion of mandatory terms and negative exclusion of deprecated vocabulary without altering the underlying model parameters.

Prior literature on constrained decoding has predominantly concentrated on structural search space modifications. Landmark frameworks such as Grid Beam Search (GBS) (Hokamp and Liu 2017) and Dynamic Beam Allocation (DBA) (Post and Vilar 2018; Hu et al. 2019) explicitly track constraint progression across discrete beam banks, dynamically allocating candidate hypotheses according to the number of satisfied lexical items. More recent extensions, such as NeuroLogic Decoding (Lu et al. 2021) and Cascaded Beam Search (CBS) (Odermatt et al. 2023), incorporate predicate logic or tiered candidate pools to handle complex conjunctive and disjunctive constraints.

Despite their theoretical guarantees, structural beam search modifications present substantial practical drawbacks in real-world deployments. Partitioning the beam space into parallel constraint banks introduces significant computational overhead, memory fragmentation, and complex state bookkeeping, frequently slowing down generation by factors of two to five relative to standard beam search. Furthermore, structural algorithms require customized decoding loops that are difficult to integrate with standardized production inference runtimes and modular wrappers such as the HuggingFace LogitsProcessor interface (Wolf et al. 2020). Most critically, finite-state structural allocators operate under the assumption of rigid target token sequences. When applied to morphologically rich, agglutinative languages such as Turkish, where grammatical roles are concatenated directly onto word roots as extensive suffix chains (Ofłazer 1994; Oral et al. 2024), structural matchers frequently fail to recognize inflected surface forms. In such scenarios, the decoder enters a structural failure state where constraint banks cannot close cleanly, causing the model to emit runaway word-appending artifacts at the end of generated sequences before the end-of-sequence token can be triggered.

To overcome these structural limitations, this paper investigates a lightweight, modular, and beam-search-agnostic alternative: *decoding-time logit manipulation*. Rather than altering the underlying beam search traversal, logit-level intervention operates directly upon the raw, unnormalized output logit vectors produced by the neural network at each autoregressive generation step. By dynamically adjusting the logit distribution prior to the softmax transformation and beam expansion, we guide the search trajectory toward required tokens and away from forbidden tokens while leaving the underlying beam search mechanics completely untouched.

In this work, we formalize a comprehensive taxonomy of logit-level decoding strategies spanning hard exclusion, hard inclusion with dynamic anchor scheduling, soft logit penalties, and curriculum-anchored soft rewards with multi-tier escalation ladders, alongside simultaneous combined modes. We evaluate these methods on an English $\leftrightarrow$ Turkish (EN $\leftrightarrow$ TR) translation benchmark. Turkish presents an ideal linguistic testbed due to its agglutinative morphology, vowel harmony constraints, and flexible Subject-Object-Verb (SOV) sentence structure, providing a sharp typological contrast to the analytic, Subject-Verb-Object (SVO) structure of English.

Our investigation addresses four central research questions regarding the efficacy, morphological robustness, optimization dynamics, and interpretability of logit-level constrained decoding. First, we examine whether logit-level manipulation can achieve constraint satisfaction rates comparable to structural beam-search algorithms while preserving natural translation fluency. Second,

we investigate how logit-level strategies and structural allocators behave when enforcing constraints in agglutinative target languages, assessing the utility of morphology-aware constraint expansion. Third, we characterize the optimization dynamics and hyperparameter landscapes across different steering mechanisms, explaining parameter plateaus and directional asymmetries. Finally, we quantify the token-level probability adjustments required to enforce lexical constraints across different semantic contexts.

The main contributions of this work are summarized as follows:

- (1) We develop a modular, beam-search-agnostic logit manipulation framework that enforces both hard and soft lexical constraints directly through standard HuggingFace `LogitsProcessor` interfaces without modifying model weights or beam search internals.
- (2) We introduce a dynamic anchor pressure schedule and a multi-tier curriculum escalation mechanism to prevent early-token grammatical disruption, paired with morphology-aware constraint expansion to handle agglutinative target suffix structures.
- (3) We present a comprehensive bidirectional evaluation on English $\leftrightarrow$ Turkish machine translation benchmarking six constrained decoding strategies against the HuggingFace Dynamic Beam Allocation baseline across a bilingual corpus of 500 sentence pairs.
- (4) We provide a detailed mechanistic characterization of structural beam inflation in DBA on agglutinative targets, the logit squeeze effect in simultaneous multi-constraint decoding, threshold suppression plateaus in soft penalties, and token-level interpretability dynamics.

The remainder of this article is structured as follows. Section 2 reviews related work across structural constrained decoding, training-time terminology integration, logit manipulation, and Turkish morphological processing. Section 3 presents our methodology, formal mathematical formulations, and algorithmic specifications. Section 4 details the dataset mining pipeline, constraint taxonomy, baseline configurations, hyperparameter optimization framework, and evaluation protocol. Section 5 presents aggregate empirical results, in-depth hyperparameter landscape analyses, token-level interpretability findings, and an extensive failure taxonomy. Section 6 walks through qualitative linguistic case studies. Section 7 discusses methodological limitations and future directions, and Section 8 concludes the paper.

2. Related Work and Theoretical Foundations

Lexically constrained text generation in neural machine translation has evolved along three primary paradigms: structural search-space modifications, training-time terminology injection, and decoding-time logit steering. This section reviews these paradigms and contextualizes the challenges posed by morphologically rich languages.

2.1 Structural Search Space Modifications

The foundational approach to lexically constrained decoding modifies the underlying graph traversal of beam search to track constraint fulfillment explicitly. Anderson et al. (2017) introduced Finite State Acceptor (FSA) guidance in image captioning, maintaining separate beam states for each FSA state. Building upon this, Hokamp and Liu (2017) introduced Grid Beam Search (GBS) for neural machine translation. GBS arranges beam hypotheses into a two-dimensional grid where the horizontal axis corresponds to generated sequence length and the vertical axis corresponds to the number of satisfied lexical constraints. While GBS guarantees constraint satisfaction, its computational complexity scales linearly with the number of constraints C and beam size B , resulting in a time complexity of $\mathcal{O}(B \times C)$ per decoding step, which leads to substantial latency degradation.

To mitigate this computational bottleneck, Post and Vilar (2018) proposed Dynamic Beam Allocation (DBA). Instead of maintaining independent grids, DBA dynamically partitions a single fixed-size beam pool across multiple discrete banks, where Bank k holds hypotheses that have satisfied exactly k constraints. By sharing a global beam budget of size B , DBA reduces decoding time by an order of magnitude relative to GBS. Hu et al. (2019) subsequently extended DBA to support multi-token phrasal constraints and disjunctive sets through trie-based prefix matching.

Subsequent research has expanded structural search through logical and cascading formulations. Lu et al. (2021) introduced NeuroLogic Decoding, which frames lexically constrained generation as satisfying complex predicates expressed in Conjunctive Normal Form (CNF). NeuroLogic guides beam generation using an aggressive lookahead heuristic and candidate pruning. Miao et al. (2019) explored sampling-based constrained generation using Metropolis-Hastings Markov Chain Monte Carlo (MCMC) sampling with candidate replacements. In a recent development, Odermatt et al. (2023) proposed Cascaded Beam Search (CBS), which partitions beam exploration into tiered priority levels, enforcing terminology while attempting to prune redundant search paths.

Despite their algorithmic sophistication, structural methods share fundamental drawbacks: they require non-trivial modifications to the beam search algorithm, incur substantial memory allocation overhead, and assume that constraints appear as exact, contiguous subword sequences. When target languages employ complex affixation, structural matching easily stalls, as explored in Section 5.3.

2.2 Training-Time Terminology Integration

An alternative paradigm bakes lexical constraints directly into model parameters or input representations during training. Dinu et al. (2019) pioneered terminology-aware training by appending inline XML-style constraint annotations to source sentences during fine-tuning. This encourages the encoder-decoder attention mechanism to copy or align the specified target lemma into the generated translation.

Subsequent variations have explored synthetic terminology augmentation and tag-based conditioning on large parallel corpora (Alam et al. 2021; Semenov et al. 2023). While training-time approaches avoid decoding-time latency overhead and allow the neural network to naturally integrate target syntax around specified terms, they suffer from critical operational limitations. Specifically, models trained with terminology markup struggle when confronted with unseen vocabulary, morphologically novel words, or complex multi-word expressions not encountered during training. Furthermore, inducing high copy biases on synthetic tags often degrades unconstrained sentence fluency and can lead to hallucinated repetition when multiple constraints overlap (Baek et al. 2023). Retraining or fine-tuning models for enterprise environments with frequently changing customer glossaries also introduces prohibitive engineering overhead.

Evaluating the delicate equilibrium between terminology adherence and translation quality remains a central challenge in machine translation assessment (Chatzikoumi 2020). Consequently, inference-time decoding methods remain essential for robust, zero-shot lexical enforcement.

2.3 Decoding-Time Logit Manipulation and Steering

Logit manipulation represents a lightweight, plug-and-play paradigm that steers autoregressive language generation by directly modifying output probability distributions prior to sampling or beam selection. This approach requires no parameter updates and operates transparently with any sequence-to-sequence model exposing a logits transformation hook.

Pascual et al. (2021) introduced Plug-and-Play Language Models (PPLM) for controlled text generation, computing gradient updates from keyword-matching loss functions to shift output

token distributions toward target semantic clusters. Sha (2020) extended gradient-guided steering to unsupervised lexically constrained generation. Beyond gradient steering, direct logit modification has been widely used for negative constraints, such as nucleus filtering (Holtzman et al. 2020) and forbidden n-gram suppression (Freitag and Al-Onaizan 2017).

In constrained machine translation, logit manipulation is particularly attractive because it operates in $\mathcal{O}(1)$ time relative to beam size and can be seamlessly integrated into standardized inference pipelines such as the HuggingFace LogitsProcessor interface (Wolf et al. 2020). However, prior attempts at direct logit boosting often suffered from severe grammatical disruption: naively increasing the logit of a target word from the very first decoding step forces the model to emit the word prematurely, breaking subject-verb agreement and sentence structure. Addressing this disruption requires sophisticated dynamic pressure scheduling and contextual anchoring, which form the core of our proposed methodology.

2.4 Morphological Richness and Agglutinative Alignment

Morphologically rich languages present acute challenges for neural machine translation and constrained decoding. Modern NMT architectures tokenize text using subword segmentation algorithms such as Byte-Pair Encoding (BPE) (Sennrich et al. 2016) or WordPiece. In analytic languages like English, words typically correspond to one or two subword tokens, and grammatical relations are encoded via separate function words such as prepositions, auxiliary verbs, and articles.

In sharp contrast, agglutinative languages such as Turkish, Finnish, and Hungarian concatenate multiple grammatical morphemes directly onto the root lemma (Oflazer 1994; Oral et al. 2024). For example, the Turkish root noun *ev* (“house”) can produce complex inflected surface forms such as *evlerimizden* (“from our houses”), decomposed into the root *ev*, plural suffix *-ler*, 1st-person plural possessive suffix *-imiz*, and ablative case marker *-den*.

When a user specifies a lexical constraint, it is almost universally provided in its dictionary citation form (lemma), such as *hekim* (“physician”). However, in context, the target sentence may demand an inflected accusative form (*hekimi*), dative form (*hekime*), or genitive form (*hekimin*). If a constrained decoding algorithm enforces exact-string matching on the bare lemma, two failure modes emerge. First, the model is forced to emit the uninflected lemma in a syntactic position requiring case marking, producing ungrammatical Turkish. Second, structural allocators like DBA fail to recognize that the inflected form satisfies the semantic intent, keeping constraint banks open and causing runaway generation.

Jon et al. (2021) addressed morphological agreement in Czech constrained NMT by supplying lemmatized constraints at training time. In our purely inference-time setting, we address Turkish agglutination through morphology-aware subword expansion and calibrated boundary penalties, detailed in Section 3.2.

3. Methodology and Algorithmic Design

This section formalizes our logit-level constrained decoding framework. We describe the end-to-end architecture, mathematical formulations of individual processors, dynamic anchor scheduling, curriculum escalation ladders, and morphological boundary enforcement.

3.1 End-to-End System Architecture

Our framework operates within the standard autoregressive sequence-to-sequence generation loop of the Helsinki-NLP OPUS-MT transformer architecture (Tiedemann and Thottingal 2020; Tiedemann et al. 2024). Let $\mathbf{x} = (x_1, x_2, \dots, x_{L_{\text{src}}})$ denote the tokenized source sentence of length

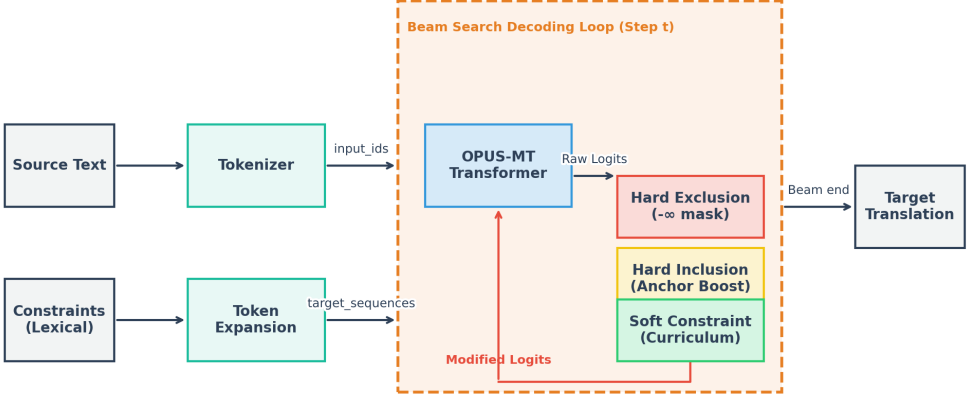


Figure 1. System architecture of the proposed logit manipulation framework. At each autoregressive decoding step t , active `LogitsProcessor` instances intercept the unnormalized output logits $\mathbf{l}_t \in \mathbb{R}^{|\mathcal{V}|}$ produced by the MarianMT decoder, applying dynamic anchor pressure, soft rewards, or hard masks before beam expansion and hypothesis pruning.

L_{src} , and let $\mathbf{y}_{<t} = (y_1, y_2, \dots, y_{t-1})$ denote the partially generated target sequence up to decoding step $t - 1$.

At step t , the transformer decoder produces a vector of unnormalized output logits over the target vocabulary $\mathcal{V}$:

$$\mathbf{l}_t = f_{\text{decoder}}(\mathbf{x}, \mathbf{y}_{<t}) \in \mathbb{R}^{|\mathcal{V}|}. \quad (1)$$

In unconstrained decoding, the logit vector $\mathbf{l}_t$ is converted to a probability distribution via the softmax function:

$$P(y_t = v \mid \mathbf{x}, \mathbf{y}_{<t}) = \frac{\exp(l_t^{(v)})}{\sum_{v' \in \mathcal{V}} \exp(l_t^{(v')})}, \quad (2)$$

and beam search maintains the top B highest-scoring hypotheses according to cumulative log-probabilities.

Under our proposed framework, active `LogitsProcessor` modules intercept $\mathbf{l}_t$ and apply additive or replacement transformations $\Delta \mathbf{l}_t$ prior to softmax normalization:

$$\tilde{\mathbf{l}}_t = \mathbf{l}_t + \Delta \mathbf{l}_t(\mathbf{y}_{<t}, \mathbf{x}, \mathcal{C}), \quad (3)$$

where $\mathcal{C} = \{\mathcal{F}, \mathcal{R}\}$ represents the set of forbidden exclusion words $\mathcal{F}$ and required inclusion words $\mathcal{R}$. The entire decoding pipeline is illustrated in Figure 1.

3.2 Morphology-Aware Constraint Expansion

To resolve the tension between dictionary citation lemmas and inflected target surface forms in agglutinative languages, we implement an automated subword constraint expansion preprocessor.

Given a target constraint word $w \in \mathcal{R}$, the processor constructs a set of plausible surface-form variants $\mathcal{S}(w)$ by generating orthographic, casing, and inflectional variants. For Turkish targets, we systematically concatenate common nominal case markers (accusative *-i/-ı/-u/-ü*, dative *-e/-a*, locative *-de/-da/-te/-ta*, ablative *-den/-dan/-ten/-tan*, genitive *-in/-ın/-un/-ün*), possessive suffixes (*-im, -in, -i, -imiz, -iniz, -leri*), and plural markers (*-ler/-lar*), respecting Turkish dotted/dotless *İ/i* and *I/ı* distinctions.

Each surface form $s \in \mathcal{S}(w)$ is tokenized into its constituent subword token ID sequence using the target model tokenizer:

$$\mathbf{seq}(s) = \text{Tokenizer}(s) = (t_1, t_2, \dots, t_m). \quad (4)$$

This yields a master collection of acceptable token sequences $\mathcal{T}(w) = \{\mathbf{seq}(s) \mid s \in \mathcal{S}(w)\}$. The constraint word w is considered satisfied in hypothesis $\mathbf{y}_{<t}$ if any subword sequence in $\mathcal{T}(w)$ appears as a contiguous subsequence within $\mathbf{y}_{<t}$.

3.3 Hard Exclusion Processor

Hard exclusion guarantees that forbidden vocabulary tokens never appear in generated translations. Let $\mathcal{F}_{\text{ids}} \subset \mathcal{V}$ denote the complete set of subword token IDs associated with all forbidden words in $\mathcal{F}$.

At each decoding step t , the processor sets the logits of all forbidden token IDs to negative infinity:

$$\tilde{l}_t^{(i)} = -\infty \quad \forall i \in \mathcal{F}_{\text{ids}}. \quad (5)$$

Because $\exp(-\infty) = 0$, the corresponding softmax probabilities strictly vanish:

$$P(y_t = i \mid \mathbf{x}, \mathbf{y}_{<t}) = 0 \quad \forall i \in \mathcal{F}_{\text{ids}}. \quad (6)$$

This operation executes in $\mathcal{O}(|\mathcal{F}_{\text{ids}}|)$ time via direct tensor indexing on GPU memory, incurring negligible computational overhead while guaranteeing mathematical zero leakage of forbidden tokens.

3.4 Hard Inclusion with Dynamic Anchor Scheduling

Enforcing positive lexical inclusion is fundamentally more complex than exclusion. Naive logit boosting forces the target token into early sentence positions regardless of grammatical context, leading to syntactic incoherence and hallucinated repetitions. To prevent early-token disruption, we introduce a *Dynamic Anchor Schedule*.

Let $\mathcal{R}$ denote the set of required constraint words, and let $\mathcal{T}(w)$ be the set of valid subword sequences for word $w \in \mathcal{R}$. At decoding step t , for each beam hypothesis $b \in \{1, \dots, B\}$, the processor evaluates which constraint groups remain unsatisfied.

For an unsatisfied constraint sequence $\mathbf{seq} = (u_1, u_2, \dots, u_m)$, the processor determines whether hypothesis b has already matched a prefix of length $k < m$ such that $(y_{t-k}, \dots, y_{t-1}) = (u_1, \dots, u_k)$. If a prefix of length $k \geq 1$ has been matched, the next token u_{k+1} is a continuation token. To ensure uninterrupted sequence completion, the processor applies an authoritative continuation boost:

$$\tilde{l}_{t,b}^{(u_{k+1})} = \max_{v \in \mathcal{V}} l_{t,b}^{(v)} + \beta_{\text{boost}}, \quad (7)$$

guaranteeing that the continuation token wins beam selection at step t .

Conversely, if no prefix has been matched, the root token u_1 must be placed organically. The processor computes the boost δ_t based on three contextual signals: an early-token grace period τ , a sweet-rank candidate buffer β_{sweet} , and a dynamic linear anchor baseline.

During the first τ decoding steps ($t \leq \tau$), no boost is applied ($\delta_t = 0$), allowing the model to establish natural sentence prefixes such as subject noun phrases. If the natural rank of token u_1 in beam b is already within the top R_{sweet} candidates, the token is deemed naturally probable, and only a modest buffer β_{sweet} is added:

$$\tilde{l}_{t,b}^{(u_1)} = l_{t,b}^{(u_1)} + \beta_{\text{sweet}}. \quad (8)$$

If the token is buried with a rank exceeding R_{sweet} , the processor pulls its logit up to a dynamic target anchor baseline that scales linearly with sentence completion progress ρ_t :

$$\rho_t = \min\left(1.0, \frac{t}{0.8 \cdot L_{\text{src}}}\right), \quad (9)$$

$$\text{Anchor}_t = \max_{v \in \mathcal{V}} l_{t,b}^{(v)} + A_{\text{start}} + A_{\text{range}} \cdot \rho_t, \quad (10)$$

$$\delta_t = \max\left(0, \text{Anchor}_t - l_{t,b}^{(u_1)}\right), \quad (11)$$

$$\tilde{l}_{t,b}^{(u_1)} = l_{t,b}^{(u_1)} + \delta_t. \quad (12)$$

3.4.1 EOS Blocking and Babbling Safety Valve

To prevent the model from prematurely generating the end-of-sequence token ($\langle \text{eos} \rangle$) while required constraints remain pending, the processor penalizes the logit of the $\langle \text{eos} \rangle$ token:

$$\tilde{l}_{t,b}^{(\text{eos})} = l_{t,b}^{(\text{eos})} + \gamma_{\text{eos}}(t), \quad (13)$$

where $\gamma_{\text{eos}}(t) = -50.0$ when $t \leq L_{\text{src}}$, and decays to -20.0 when $t > L_{\text{src}}$.

To prevent infinite generation loops or runaway babbling when a constraint cannot be placed grammatically, an emergency safety valve activates when sequence length exceeds $1.5 \times L_{\text{src}}$. Beyond this threshold, EOS blocking is disabled, allowing the decoder to terminate gracefully.

3.4.2 Morphological Boundary Suffix Penalty

When a multi-token constraint word completes at step t , the model may attempt to immediately affix unconstrained grammatical morphemes that alter the root’s meaning. To enforce clean lexical boundaries, we construct a boundary mask $\mathbf{M}_{\text{boundary}} \in \{0, 1\}^{|\mathcal{V}|}$ indicating tokens that begin a new word (such as tokens with the leading subword whitespace marker). At the step immediately following constraint completion, non-boundary tokens receive a suffix penalty $\gamma_{\text{suffix}} < 0$:

$$\tilde{l}_{t+1,b}^{(v)} = l_{t+1,b}^{(v)} + \gamma_{\text{suffix}} \quad \forall v \text{ where } M_{\text{boundary}}^{(v)} = 0. \quad (14)$$

3.5 Soft Constraint Processing and Curriculum Anchoring

Soft constraints modify token distributions continuously without imposing absolute mathematical guarantees, making them ideal for stylistic guidance, preferred terminology, and soft bias injection.

3.5.1 Soft Logit Penalty

For each penalty word in $\mathcal{P}$, the processor adds a constant negative bias $\lambda_{\text{pen}} < 0$ to its associated token IDs $\mathcal{P}_{\text{ids}}$:

$$\tilde{l}_t^{(p)} = l_t^{(p)} + \lambda_{\text{pen}} \quad \forall p \in \mathcal{P}_{\text{ids}}. \quad (15)$$

This suppresses undesirable tokens while allowing the model to override the penalty if contextual probability is overwhelmingly high.

3.5.2 Curriculum-Anchored Soft Reward

For pending reward words $\mathcal{R}_{\text{soft}}$, the processor employs a curriculum-inspired dynamic growth schedule (Bengio et al. 2009). The reward strength grows monotonically with the number of active generation steps η_t during which the constraint has remained unsatisfied:

$$\lambda_{\text{eff}}(t) = \min(\lambda_{\text{max}}, \lambda_{\text{rew}} \cdot (1 + c \cdot \eta_t)), \quad (16)$$

where λ_{rew} is the base reward, c is the curriculum growth rate, and λ_{max} caps the maximum reward pressure. To avoid early disruption, reward application is suppressed for the initial three decoding steps ($t \leq 3$).

Rather than naively adding λ_{eff} directly to the raw logit, the processor applies target baseline anchoring:

$$\text{Baseline}_t = \max_{v \in \mathcal{V}} l_{t,b}^{(v)} + \theta_{\text{offset}} + \lambda_{\text{eff}}(t). \quad (17)$$

If the current logit $l_{t,b}^{(r)}$ is below Baseline_t , it is pulled up to the baseline. If the token is already above the baseline (i.e., organically probable), it receives a small contextual nudge v :

$$\tilde{l}_{t,b}^{(r)} = \begin{cases} \text{Baseline}_t & \text{if } l_{t,b}^{(r)} < \text{Baseline}_t \\ l_{t,b}^{(r)} + v & \text{otherwise.} \end{cases} \quad (18)$$

3.6 Multi-Tier Escalation Ladder

To combine the high fluency of soft curriculum rewards with the absolute satisfaction guarantee of hard constraints, we implement a three-tier escalation ladder in `soft_reward_only`.

In the first tier (Passive Soft Curriculum), the sentence is decoded using curriculum-anchored soft reward with optimized parameters. If all required words appear in the output, the translation is returned immediately in a single generation pass. If any required words remain missing, a second tier (Targeted Soft Boost) executes, setting the reward strength to λ_{max} exclusively for the missing words. If required words still remain unsatisfied after the second pass, the system escalates to full Hard Inclusion with dynamic anchor scheduling as a third-tier fallback. This multi-tier ladder ensures that easy sentences are translated with minimal interference and maximum BLEU preservation, while challenging sentences receive escalated pressure to achieve near-100% constraint satisfaction.

3.7 Combined Decoding and the Logit Squeeze Phenomenon

In practical enterprise deployment, translation systems must frequently enforce positive and negative constraints simultaneously. The Hard Combined mode executes `HardExclusionProcessor` and `HardInclusionProcessor` in a single pipeline, running exclusion first so that masked tokens are excluded from max-logit anchor calculations. The Soft Combined mode executes soft penalty and soft curriculum reward simultaneously within `SoftConstraintProcessor`, followed by a soft-boost retry pass if reward tokens remain missing.

When positive and negative constraints operate concurrently in the same sentence, they can induce a *logit squeeze effect*: suppressing multiple baseline candidate tokens while simultaneously boosting out-of-distribution constraint tokens compresses the effective probability mass into a narrow subset of vocabulary. This non-linear interaction and its empirical impact on translation quality are analyzed in Section 5.5.

4. Dataset Construction and Experimental Setup

This section describes our dataset mining pipeline, constraint taxonomy, baseline configurations, hyperparameter optimization framework, and evaluation protocol.

4.1 Dataset Mining Pipeline and Constraint Taxonomy

To evaluate constrained decoding rigorously across diverse linguistic settings, we constructed a bilingual evaluation corpus of 500 sentence pairs (250 English→Turkish and 250 Turkish→English) derived from the Tatoeba / FLORES-200 parallel corpus challenge (Tiedemann 2020).

Constraints were extracted using an automated reference-difference mining pipeline. First, for each source sentence $\mathbf{x}$, we generate an unconstrained translation $\mathbf{y}_{\text{base}}$ using the baseline OPUS-MT model with standard 16-beam search. Both the human reference $\mathbf{y}_{\text{ref}}$ and the baseline output $\mathbf{y}_{\text{base}}$ are lowercased and stripped of punctuation. Language-specific stopword lists (comprising high-frequency particles, pronouns, auxiliaries, and conjunctions in Turkish and English) and short tokens under four characters are removed to isolate substantive content words.

To avoid self-contradictory constraint specifications, such as requiring the word *doktor* while forbidding its inflected variant *doktoru*, we apply a morphological overlap filter. Two words are deemed morphologically related if they share a common three-character prefix:

$$\text{Overlap}(w_1, w_2) = \mathbb{I}(\text{prefix}_3(w_1) = \text{prefix}_3(w_2)). \quad (19)$$

Tokens exhibiting morphological overlap with required words are strictly excluded from the forbidden constraint set.

The sentence pairs are split evenly into two difficulty tiers based on the baseline model’s intrinsic lexical distribution. In the Easy tier, the required constraint word $w \in (\mathbf{y}_{\text{ref}} \cap \mathbf{y}_{\text{base}})$ is already produced by the baseline model in unconstrained generation. This tier evaluates whether the constrained decoding mechanism preserves natural fluency without introducing spurious distortions or duplicate insertions. In the Hard tier, the required constraint word $w \in (\mathbf{y}_{\text{ref}} \setminus \mathbf{y}_{\text{base}})$ is completely absent from the baseline translation, evaluating the algorithm’s capability to steer the beam search trajectory into low-probability semantic subspaces and override baseline lexical choices.

The 500 test sentences span a rich taxonomy of linguistic domains and syntactic challenges, including technical and scientific terminology (e.g., *yapay zeka*, *kuantum bilgisayarlar*, *algoritma*), biomedical and clinical vocabulary (e.g., *hekim*, *ameliyat/operasyon*), economic and legal terms (e.g., *mevzuat*, *merkez bankası*, *faiz oranları*), multi-word phrasal compounds (e.g., *uzay istasyonu*, *küresel ısınma*), and agglutinative suffix chains. The corpus is partitioned into a dedicated Hyperparameter Optimization (HPO) development set and a strictly held-out final evaluation set.

4.2 HuggingFace Dynamic Beam Allocation (DBA) Baseline

As our primary competitive baseline, we implement the state-of-the-art HuggingFace Transformers Dynamic Beam Allocation (DBA) algorithm (Post and Vilar 2018; Wolf et al. 2020).

In HuggingFace Transformers, lexical inclusion constraints are enforced via the `DisjunctiveConstraint` class, which encapsulates multi-token sequences within a `DisjunctiveTrie`. However, the standard `DisjunctiveTrie` implementation contains an algorithmic vulnerability: if any token sequence is a strict prefix of another sequence within the same disjunctive group, the trie construction throws a fatal exception. To ensure robust evaluation, we developed a prefix-subset filtering algorithm:

$$\mathcal{T}_{\text{filtered}} = \{s \in \mathcal{T} \mid \neg \exists s' \in \mathcal{T} \text{ such that } s' \text{ is a strict prefix of } s\}. \quad (20)$$

This guarantees clean trie compilation across all morphological variants. Forbidden exclusion words in the DBA baseline are enforced via `bad_words_ids`, mapped to flat token ID lists.

4.3 Hyperparameter Optimization Framework

Hyperparameters for all decoding strategies and the DBA baseline were systematically optimized using Optuna (Akiba et al. 2019) with the Tree-structured Parzen Estimator (TPE) Bayesian sampling algorithm. Optimization was conducted independently for each translation direction on the dedicated HPO set.

To balance constraint satisfaction against translation fluency and prevent output length distortion, we formulated a composite objective function:

$$\mathcal{L}_{\text{obj}} = (S \times 200) + \overline{\text{BLEU}} - \mathcal{P}_{\text{len}}, \quad (21)$$

where $S \in [0, 1]$ represents the constraint satisfaction rate, $\overline{\text{BLEU}}$ is the sentence-level BLEU score computed relative to the unconstrained baseline output, and $\mathcal{P}_{\text{len}}$ is a quadratic length inflation penalty:

$$\mathcal{P}_{\text{len}} = \max(0, R_{\text{len}} - \theta)^2 \times 150,000, \quad (22)$$

where $R_{\text{len}} = L_{\text{hyp}}/L_{\text{base}}$ is the average output length ratio, and θ is a tolerance threshold ($\theta = 1.05$ for logit-level modes; $\theta = 1.03$ for DBA). The quadratic multiplier of 150,000 imposes severe penalties on candidates that achieve high satisfaction by artificially padding or appending words: an average length ratio of $R_{\text{len}} = 1.20$ contributes a massive penalty of $-3,375$, immediately disqualifying degenerate candidates.

Optimization trial budgets were allocated based on search space dimensionality: 50 trials per direction for Hard Inclusion, Hard Combined, and Soft Combined (spanning 5 to 7 continuous/discrete parameters); 30 trials per direction for HuggingFace DBA; 30 trials per direction for Soft Reward; and 10 trials per direction for Soft Penalty. The complete set of optimized hyperparameters is summarized in Table 1.

Table 1. Optimized hyperparameters discovered by Optuna TPE per translation direction on dedicated HPO splits. Parameters marked with $\dagger$ exhibit significant cross-lingual directional asymmetry.

Strategy Group	Parameter Name	EN→TR	TR→EN
Hard Inclusion	Early-token grace $\tau^\dagger$	0	5
	Sweet-rank threshold $R_{\text{sweet}}^\dagger$	557	293
	Sweet-rank buffer $\beta_{\text{sweet}}^\dagger$	12.27	5.86
	Anchor start A_{start}	-6.55	-7.50
	Anchor range A_{range}	10.31	9.83
	Continuation boost β_{boost}	5.74	7.18
	Suffix penalty $\gamma_{\text{suffix}}^\dagger$	-21.92	-9.34
Soft Reward	Base reward λ_{rew}	2.03	2.01
	Reward ceiling λ_{max}	15.37	19.04
	Curriculum growth rate $c^\dagger$	2.96	0.17
	Anchor offset $\theta_{\text{offset}}^\dagger$	-23.89	-28.74
Soft Penalty	Penalty magnitude λ_{pen}	-98.37	-118.52
HuggingFace DBA	Beam size $B^\dagger$	7	5
	Length penalty $\alpha^\dagger$	+0.01	-4.61
	Repetition penalty r	1.03	1.20

The operational roles of these hyperparameter groups reflect distinct decoding mechanics. Within Hard Inclusion, the early-token grace period τ defines the initial step window during

which logit boosting is withheld. The sweet-rank threshold R_{sweet} establishes the candidate cutoff for organic consideration, where high-ranking tokens receive a modest buffer β_{sweet} rather than full anchor pressure. For tokens ranked beyond R_{sweet} , the dynamic anchor parameters A_{start} and A_{range} dictate the initial offset and total upward logit climb across sentence progression. Once an initial root token is placed, the continuation boost β_{boost} enforces uninterrupted generation of multi-token subword spans, while the suffix penalty γ_{suffix} suppresses immediate unconstrained morpheme attachment at morphological boundaries.

In Soft Reward decoding, the base reward λ_{rew} and curriculum rate c regulate the monotonic growth of reward pressure per active generation step, bounded by the maximum ceiling λ_{max} . The anchor offset θ_{offset} establishes the passive safety-net baseline relative to the maximum logit, ensuring minimal distortion of organic generation paths. For Soft Penalty, the penalty magnitude λ_{pen} controls negative logit bias. For the HuggingFace DBA baseline, beam size B , length penalty α , and repetition penalty r jointly govern hypothesis bank capacity, sequence termination thresholds, and n-gram penalty scaling.

4.4 Evaluation Metrics and Reproducibility Protocol

To evaluate systems comprehensively, we report six distinct evaluation metrics. Constraint Satisfaction Rate (S) measures the percentage of test sentences satisfying all lexical inclusion and exclusion constraints, verified via a morphology-aware matcher accepting valid Turkish inflectional suffixes. Baseline BLEU ($\text{BLEU}_{\text{base}}$) computes sentence BLEU against unconstrained baseline outputs to quantify trajectory preservation. Reference BLEU (BLEU_{ref}) (Papineni et al. 2002) and ChrF (ChrF_{ref}) (Popović 2015) measure absolute quality against human references. Output Length Ratio ($R_{\text{len}} = L_{\text{hyp}}/L_{\text{base}}$) serves as a diagnostic for word-appending artifacts. Finally, Decoding Latency (ms) and generation pass counts measure computational efficiency.

All experiments were executed across three distinct random seeds ($\text{Seed} \in \{42, 123, 7\}$) on an NVIDIA GPU with CUDA acceleration in no-gradient inference mode. Reported values represent the tri-seed arithmetic mean.

5. Empirical Evaluation and Hyperparameter Dynamics

This section presents aggregate empirical results, in-depth hyperparameter landscape analyses, a mechanistic dissection of DBA structural beam failure, and quantitative token-level interpretability findings.

5.1 Aggregate Performance Comparison

Tables 2 and 3 present aggregate evaluation metrics for English→Turkish (EN→TR) and Turkish→English (TR→EN) translation, respectively. Figures 2, 4, and 5 provide visual comparisons across translation quality, length ratio, and decoding latency.

5.2 Hyperparameter Optimization Landscape, Plateaus, and Directional Asymmetries

An exhaustive analysis of the Optuna optimization trials reveals distinct structural characteristics across parameter landscapes, providing critical insights for practitioners.

Table 2. Aggregate empirical results for English to Turkish translation (EN→TR, $N_{\text{excl}} = 163$, $N_{\text{incl}} = 250$). Averaged over 3 random seeds (42, 123, 7).

Decoding Strategy	N	Satisfaction	BLEU _{base}	BLEU _{ref}	ChrF _{ref}	Length Ratio	Latency (ms)	Passes
Unconstrained Baseline	250	–	100.00	40.97	67.04	1.000	96.9	1.00
Hard Exclusion	163	98.2%	35.20	35.59	58.84	1.009	165.2	1.00
Hard Inclusion (Dynamic Anchor)	250	99.6%	50.49	31.11	65.88	1.063	943.3	1.00
Hard Combined (Simultaneous)	250	91.2%	32.53	34.98	65.04	1.114	1082.6	1.00
Soft Penalty Only	163	98.2%	35.20	35.59	58.84	1.009	167.9	1.00
Soft Reward (Curriculum Escalation)	250	99.2%	72.32	40.98	71.31	1.062	838.0	1.79
Soft Combined	250	62.0%	57.65	47.62	68.53	1.007	386.6	1.37
HuggingFace DBA Baseline	250	97.6%	31.17	34.41	62.39	2.768	2003.3	1.00

Table 3. Aggregate empirical results for Turkish to English translation (TR→EN, $N_{\text{excl}} = 142$, $N_{\text{incl}} = 247$). Averaged over 3 random seeds (42, 123, 7).

Decoding Strategy	N	Satisfaction	BLEU _{base}	BLEU _{ref}	ChrF _{ref}	Length Ratio	Latency (ms)	Passes
Unconstrained Baseline	250	–	100.00	49.55	66.99	1.000	101.2	1.00
Hard Exclusion	142	99.3%	44.74	40.58	61.52	1.016	188.4	1.00
Hard Inclusion (Dynamic Anchor)	247	99.2%	73.93	44.86	69.57	1.062	239.9	1.00
Hard Combined (Simultaneous)	247	96.0%	56.55	51.07	72.66	1.050	399.4	1.00
Soft Penalty Only	142	99.3%	44.74	40.58	61.52	1.016	184.6	1.00
Soft Reward (Curriculum Escalation)	247	98.8%	81.65	49.07	71.65	1.058	314.5	1.71
Soft Combined	247	71.3%	66.52	54.75	71.62	1.013	236.8	1.32
HuggingFace DBA Baseline	250	97.2%	49.08	50.38	70.36	1.289	784.0	1.00

5.2.1 The Threshold Suppression Effect in Soft Penalty

In both translation directions, the optimization landscape for `soft_penalty` is completely flat below a critical threshold ($\lambda_{\text{pen}} \lesssim -30.0$). Trials with penalties ranging from -35.0 to -188.0 produce mathematically identical translation sequences and metric scores.

Standard beam search retains only the top B candidate sequences at each decoding step. When a penalty λ_{pen} is applied to token ID p :

$$l_t^{(p)} \leftarrow l_t^{(p)} + \lambda_{\text{pen}}. \quad (23)$$

Once λ_{pen} is sufficiently negative such that $l_t^{(p)} < \min_{k \in \text{top-}B} l_t^{(k)}$, token p falls outside the beam pruning window. Its selection probability strictly drops to zero. Making the penalty more negative does not change the rank ordering of the remaining valid vocabulary tokens. Thus, soft penalties exhibit a step-function behavior, converging in 5–10 trials without micro-tuning.

5.2.2 The Two-Regime Landscape and Plateau in Soft Reward

For `soft_reward`, the optimization landscape exhibits two distinct regimes governed by the anchor offset parameter θ_{offset} .

In the Aggressive Nudging Regime ($\theta_{\text{offset}} \in [-15.0, 0.0]$), the target baseline is set aggressively high:

$$\text{Baseline}_t = \max_{v \in \mathcal{V}} l_t^{(v)} + \theta_{\text{offset}} + \lambda_{\text{eff}}. \quad (24)$$

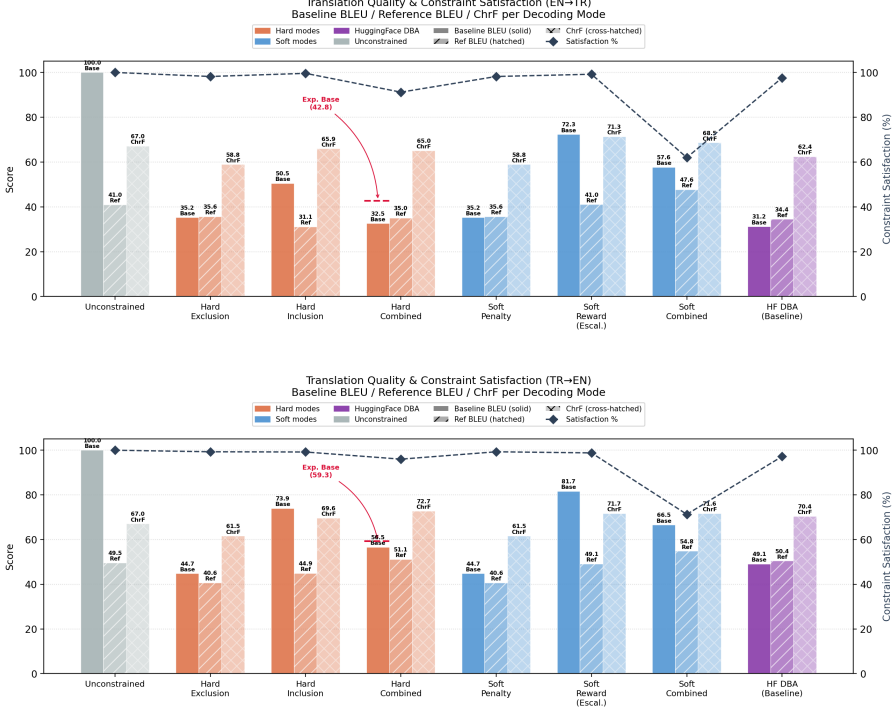


Figure 2. Comprehensive quality overview across decoding strategies: Constraint Satisfaction Rate (%), Baseline BLEU, Reference BLEU, and Reference ChrF for top panel (a) English to Turkish and bottom panel (b) Turkish to English.

The processor forces constraint tokens into early sentence positions regardless of context, disrupting syntactic fluency and dropping Baseline BLEU below 45.0.

In the Passive Safety-Net Plateau ($\theta_{\text{offset}} < -15.0$), the baseline acts as a gentle safety net. The model generates constraint tokens organically whenever possible, and the reward baseline intervenes only when a constraint is severely suppressed. Within this region, diverse combinations of reward strength λ_{rew} and offset θ_{offset} yield identical scores, forming an expansive, optimal plateau.

5.2.3 Directional Linguistic Asymmetries

A critical finding from Table 1 is the marked directional asymmetry between EN→TR and TR→EN.

Regarding the early grace period τ , in TR→EN the optimizer converged to $\tau = 5$, allowing the decoder a 5-step passive window before applying logit pressure. English SVO syntax allows initial subject and verb tokens to establish structure naturally. In contrast, EN→TR converged to $\tau = 0$ (immediate pressure). In Turkish SOV syntax, object nouns and modifying phrases frequently appear at the start of the clause; delaying pressure causes the model to commit to alternative syntactic branches from which it cannot recover without severe reordering.

Furthermore, Turkish targets require a substantially wider sweet-rank window ($R_{\text{sweet}} = 557$ in EN→TR vs. 293 in TR→EN) and larger buffer ($\beta_{\text{sweet}} = 12.27$ vs. 5.86). This reflects the higher subword vocabulary dispersion in Turkish BPE tokenization, where inflected forms disperse probability mass across hundreds of candidate subwords. Turkish targets also require an aggressive curriculum rate ($c = 2.96$) compared to English ($c = 0.17$), rapidly escalating pressure before the final finite verb is generated.

Figure 3 illustrates the resulting dynamic anchor schedules simulated across sentence progression for both translation directions.

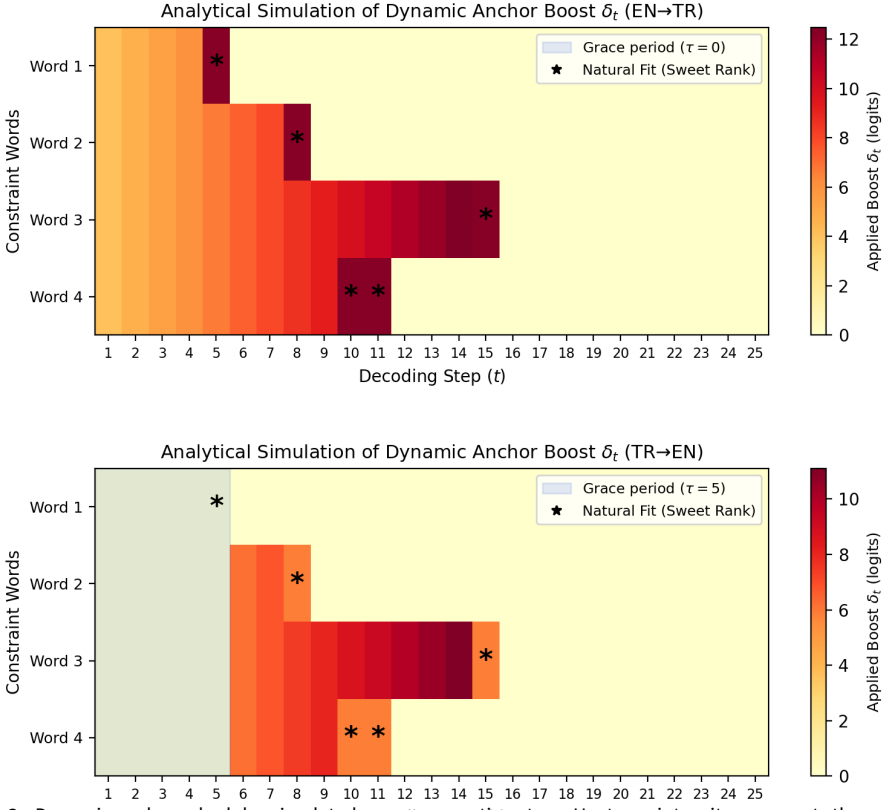


Figure 3. Dynamic anchor schedules simulated across sentence progression steps. Heatmap intensity represents the applied logit boost δ_t . Top panel (a) illustrates EN→TR parameters ($\tau = 0$, $A_{\text{start}} = -6.55$, $A_{\text{range}} = 10.31$), applying immediate pressure. Bottom panel (b) illustrates TR→EN parameters ($\tau = 5$, $A_{\text{start}} = -7.50$, $A_{\text{range}} = 9.83$), incorporating an initial 5-step passive grace period.

5.3 Architectural Anatomy of HuggingFace DBA Failure on Agglutinative Targets

The empirical evaluation uncovered a severe structural failure mode in the HuggingFace DBA baseline when translating into Turkish: DBA exhibited a catastrophic average length ratio of **2.768** in EN→TR (meaning outputs were nearly 2.8 times their natural length), accompanied by low Baseline BLEU (31.17). In TR→EN, DBA performed normally with a length ratio of 1.289 and 49.08 Baseline BLEU.

The root cause of this failure lies in the interaction between DBA’s multi-bank allocation architecture and Turkish agglutinative subword segmentation. DBA partitions beam capacity B across discrete banks $0, 1, \dots, C$, where a hypothesis cannot transition from Bank k to Bank $k + 1$ unless it matches an exact subword token sequence in the *DisjunctiveTrie*.

In Turkish, the naturally fluent translation requires an inflected suffix chain. If the disjunctive trie contains the bare lemma and selected suffix variants, but the model context favors a complex compound suffix, the generated token sequence fails to trigger the trie completion state. Because Bank C remains empty, DBA’s allocation algorithm reserves beam slots exclusively for hypotheses

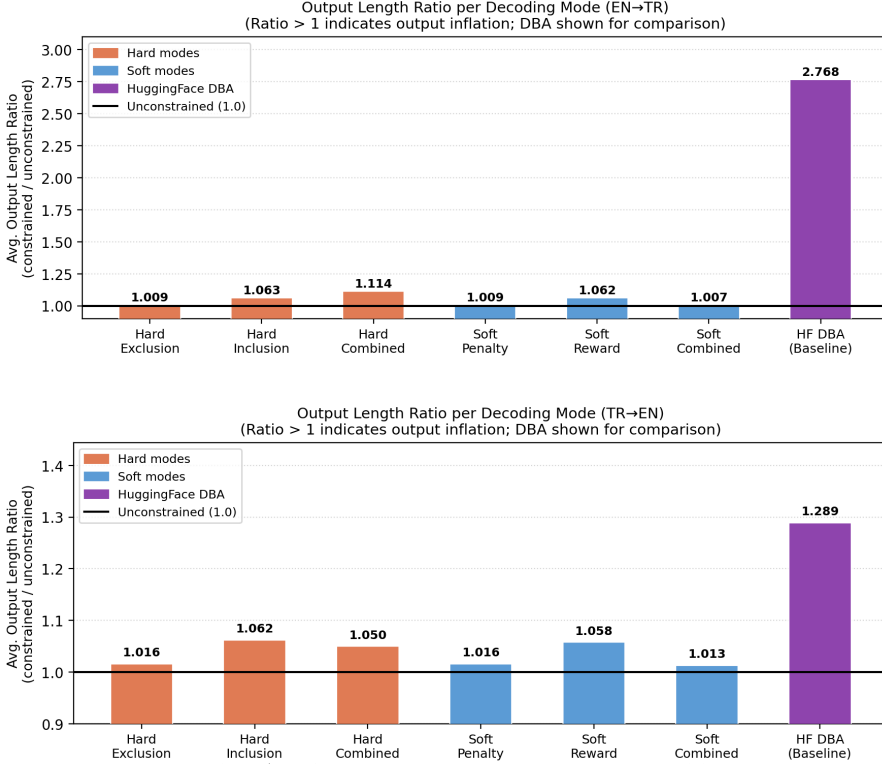


Figure 4. Output length ratio ($L_{\text{hyp}}/L_{\text{base}}$) across decoding strategies for top panel (a) EN→TR and bottom panel (b) TR→EN. HuggingFace DBA exhibits extreme length inflation in EN→TR ($2.768\times$) due to agglutinative finite-state beam traps, whereas all logit-level methods remain strictly below $1.12\times$.

attempting to fulfill the missing constraint. As the sentence reaches its natural syntactic conclusion, the model attempts to emit `<eos>`, but DBA blocks termination because Bank *C* has not been satisfied. The search is forced into a degenerative loop, repeatedly generating filler tokens, repeating punctuation, and finally appending the exact bare constraint lemma at the very end of the string before allowing `<eos>` to emit.

In contrast, our logit manipulation framework applies dynamic anchor pressure at the logit level, allowing the model’s self-attention layers to naturally integrate the required stem into an inflected surface form, achieving a clean length ratio of **1.063** in EN→TR.

5.4 Decoding Latency and Throughput Profiling

Figure 5 profiles average wall-clock decoding latency per sentence across all modes. Logit-level methods demonstrate marked computational advantages. Single-pass Hard Inclusion executes in 943.3 ms in EN→TR, representing a **2.1 $\times$ speedup** over DBA (2003.3 ms). In TR→EN, Hard Inclusion executes in 239.9 ms, achieving a **3.3 $\times$ speedup** over DBA (784.0 ms).

Regarding multi-tier escalation, soft reward with escalation averages 1.79 passes in EN→TR and 1.71 passes in TR→EN, with an average latency of 838.0 ms and 314.5 ms respectively. Approximately 60% of sentences complete in a single pass at ~ 100 ms latency, while only challenging sentences trigger retries.

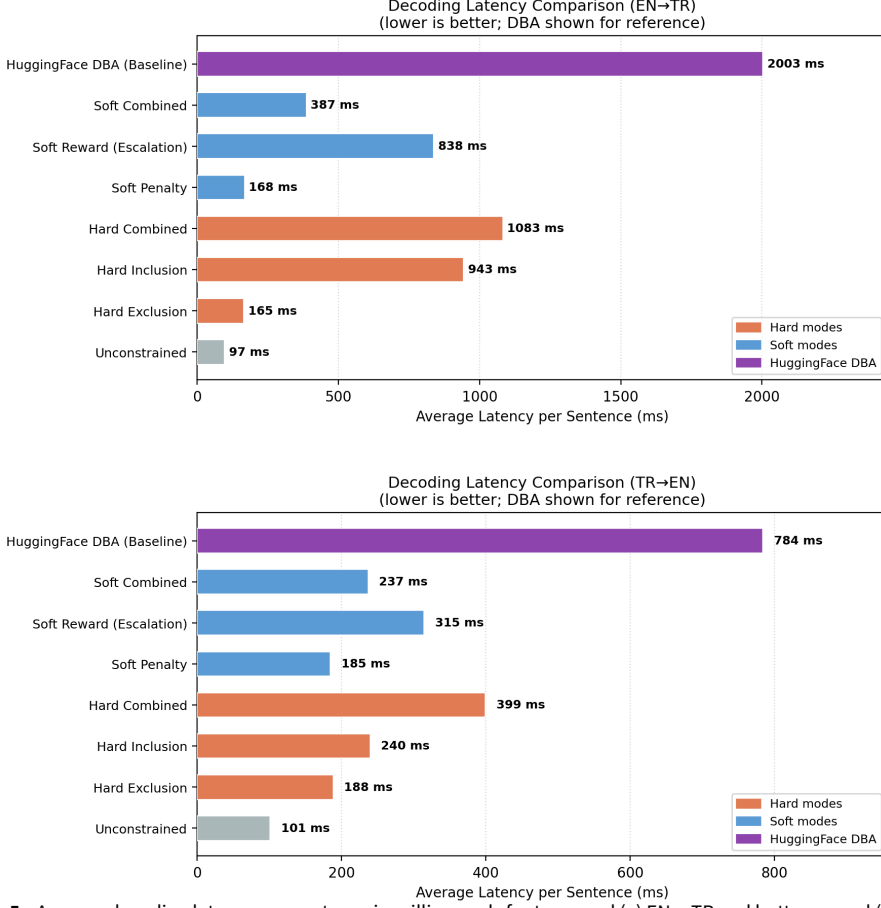


Figure 5. Average decoding latency per sentence in milliseconds for top panel (a) EN→TR and bottom panel (b) TR→EN. Hard Inclusion executes $2.1\times$ faster than DBA in EN→TR (943.3 vs. 2003.3 ms) and $3.3\times$ faster in TR→EN (239.9 vs. 784.0 ms).

5.5 Simultaneous Multi-Constraint Dynamics and the Logit Squeeze Effect

In combined modes enforcing simultaneous exclusion and inclusion, Hard Combined drops satisfaction to 91.2% in EN→TR (vs. 99.6% in Hard Inclusion) and Baseline BLEU to 32.53, while Soft Combined drops satisfaction to 62.0% in EN→TR and 71.3% in TR→EN.

When exclusion processors set multiple forbidden token logits to $-\infty$, the effective vocabulary dimension collapses. Simultaneously, soft penalties suppress related semantic synonyms. When positive anchor pressure is applied to required tokens, the model’s conditioned autoregressive prior is squeezed between competing constraints, forcing the beam search into low-probability degenerate hypotheses. This confirms that multi-constraint decoding requires decoupled, sequential scheduling rather than simultaneous uniform pressure.

5.6 Token-Level Interpretability and Probability Dynamics

To understand the internal mechanics of logit steering, we analyzed the step-level interpretability logs recorded by our framework across all 500 evaluation sentences. Table 4 summarizes pre-intervention token statistics.

Table 4. Token-level interpretability metrics before intervention. Pre-intervention rank and probability reflect the model’s natural organic consideration of constraint tokens prior to logit manipulation.

Decoding Strategy	Active Steps	Avg. Pre-Rank	Avg. Pre-Prob	Avg. $ \Delta I $ Applied
Hard Inclusion (EN→TR)	17.2	1,814.7	7.21×10^{-4}	14.75
Hard Inclusion (TR→EN)	12.4	1,248.3	1.15×10^{-3}	11.20
Soft Reward (EN→TR)	14.0	4,974.6	2.33×10^{-5}	2.00
Soft Reward (TR→EN)	10.8	3,812.1	4.18×10^{-5}	2.00
Hard Combined (EN→TR)	16.0	882.2	4.33×10^{-4}	23.49

In hard constraint cases, the target constraint token initially ranks on average between position 1,200 and 5,000 in the model’s unnormalized distribution, with softmax probabilities below 10^{-4} . Hard inclusion applies an average logit delta of +14.75 in EN→TR to elevate buried tokens into the active beam, whereas soft reward operates via modest +2.00 contextual nudges on organic candidates, escalating only when tokens remain buried.

5.7 Mechanistic Analysis of Framework Components

The design of our logit manipulation framework incorporates five distinct architectural components, each targeting a specific structural failure mode in autoregressive constrained generation:

Dynamic Anchor Progress Scheduling (ρ_t). In unconstrained decoding, the autoregressive decoder establishes grammatical structure by placing subject noun phrases, determiners, or clause-initial adverbials during early generation steps. A static, uncalibrated logit boost applied from step $t = 1$ forces low-ranking constraint tokens into early sentence positions regardless of syntactic context, leading to incoherent sentence prefixes and unnatural word order. By scaling the anchor target linearly with sentence completion progress $\rho_t = \min(1.0, t/(0.8 \cdot L_{\text{src}}))$, our framework allows the self-attention mechanism to formulate the natural sentence frame during initial steps ($t \leq \tau$) before progressively elevating the logit baseline for unsatisfied constraint tokens.

Authoritative Continuation Boosting (β_{boost}). In subword tokenization models (such as BPE and WordPiece), lexical constraints frequently decompose into multi-token sequences ($u_1, u_2, \dots, u_m$). If only the root token u_1 is boosted, the remaining continuation tokens $u_2, \dots, u_m$ must compete against the model’s unconstrained beam candidates at subsequent steps. When generating complex morphological compounds, high-frequency function words or alternative subword branches can intercept the sequence, leaving the constraint split or truncated. The authoritative continuation boost β_{boost} guarantees that once hypothesis b matches a valid prefix ($u_1, \dots, u_k$), the subsequent continuation token u_{k+1} decisively wins beam expansion, ensuring atomic and contiguous sequence completion.

Morphological Boundary Suffix Penalty (γ_{suffix}). In agglutinative target languages such as Turkish, the decoder language model head possesses a strong prior to affix trailing derivational and inflectional morphemes to completed noun roots. When a multi-token constraint sequence finishes, unconstrained suffix concatenation can alter the fundamental grammatical role or lexical meaning of the constraint. The boundary suffix penalty γ_{suffix} suppresses non-boundary subwords at the step immediately following constraint completion, enforcing a clean word boundary before allowing the model to resume natural sentence generation.

EOS Blocking and Babbling Safety Valve. Standard beam search readily terminates sequences by emitting the $\langle \text{eos} \rangle$ token when the sentence reaches a syntactically plausible conclusion, even if mandatory lexical constraints remain unplaced. The dynamic penalty $\gamma_{\text{eos}}(t)$ applies strong suppression (-50.0) during the primary generation window ($t \leq L_{\text{src}}$) and moderate suppression (-20.0) as length increases, giving the decoder the requisite step budget to integrate pending constraints. Crucially, to prevent infinite generation loops or repetitive degenerate babbling

when a constraint cannot be placed grammatically, the emergency safety valve disengages EOS suppression once sequence length exceeds $1.5 \times L_{\text{src}}$, allowing the decoder to terminate gracefully.

Multi-Tier Escalation in Soft Reward. The three-tier escalation ladder resolves the tension between unconstrained translation quality and strict constraint satisfaction. Tier 1 (Passive Soft Curriculum) applies gentle, context-aware nudges that allow the decoder to satisfy constraints organically without distorting baseline translation trajectories. Sentences that omit required terms under Tier 1 automatically escalate to Tier 2 (Targeted Soft Boost with reward strength set to the maximum ceiling λ_{max} exclusively for omitted constraints) and Tier 3 (Hard Fallback with dynamic anchor scheduling). This multi-tier design ensures that the majority of test sentences complete in a single, high-fluency pass with minimal latency, reserving aggressive intervention exclusively for challenging syntactic contexts.

5.8 Domain-Specific Linguistic and Morphosyntactic Dynamics

The curated bilingual evaluation corpus spans diverse linguistic registers and syntactic structures, revealing distinct interaction patterns between logit-level steering and target language typologies:

Biomedical and Clinical Terminology. Biomedical text frequently incorporates specialized domain terminology and loanword roots (e.g., *hekim*, *operasyon*, *semtom*, *tedavi*). In Turkish, these terms must conform to regular vowel harmony and nominal case marking (such as accusative *-i/-ı/-u/-ü* and dative *-e/-a*). When constrained decoding systems operate under rigid string-matching assumptions, they either force uninflected citation forms into inflected slots (creating ungrammatical output) or fail to detect valid inflections. Our morphology-aware subword expansion enables the decoder to attach appropriate inflectional suffixes while fully satisfying the underlying lexical constraint.

Legal and Regulatory Documentation. Legal and regulatory texts are characterized by multi-word phrasal expressions (e.g., *merkez bankası*, *faiz oranları*, *yürürlükteki mevzuat*) and rigid clause subordination. In these structures, lexical constraints must appear as strictly contiguous phrases without intervening adverbs or modifiers. Authoritative continuation boosting ensures that once the initial token of a legal phrase is selected, subsequent tokens are emitted contiguously, preserving formal legal phraseology without structural fragmentation.

Technical Terminology and Orthographic Sensitivity. Technical and computing terminology features frequent acronyms, borrowed stems, and compound nouns (e.g., *yapay zeka*, *algoritma*, *kuantum bilgisayarlar*). Turkish orthography enforces distinct uppercase/lowercase mappings for dotted *İ/i* and dotless *I/i*, which are treated as distinct subword tokens in BPE vocabularies. Preprocessing constraints through casing-aware subword expansion ensures that orthographic variants are matched consistently across both English and Turkish translation directions.

Agglutinative Morphological Chains vs. Structural Beam Deadlock. In morphologically rich, agglutinative targets, sentences exhibit complex suffix chains where multiple grammatical relations are concatenated directly onto nominal and verbal stems. This linguistic characteristic exposes the fundamental structural vulnerability of Dynamic Beam Allocation: when an inflected target form diverges from the exact subword sequence registered in the disjunctive trie, DBA’s constraint banks fail to close, trapping the search in a degenerative word-appending loop. In contrast, logit-level manipulation steers token probabilities directly at the output layer, allowing the transformer self-attention heads to seamlessly synthesize inflected surface forms while preserving natural sequence length.

Conversational and Everyday Expressions. In colloquial and general-domain sentences, syntactic templates exhibit high flexibility, and multiple valid paraphrases exist for a given source utterance. In this regime, curriculum-anchored soft reward proves particularly advantageous: by

providing modest, time-dependent probability nudges rather than hard logit overrides, the system preserves idiomatic naturalness and baseline fluency while achieving near-complete lexical adherence.

5.9 Beam Search Exploration Dynamics and Computational Complexity

A fundamental architectural advantage of logit manipulation lies in how it preserves the global beam search traversal relative to structural beam allocation algorithms:

Global vs. Partitioned Beam Budget. In Dynamic Beam Allocation (DBA), the global beam budget of size B is partitioned across $C + 1$ discrete constraint banks $(0, 1, \dots, C)$, where Bank k exclusively holds hypotheses that have satisfied exactly k constraints. Consequently, the effective candidate pool available to any individual bank shrinks to $\approx \lfloor B/(C + 1) \rfloor$. When decoding with typical parameters ($B = 5$ and $C = 2$), each active bank retains only 1 or 2 hypotheses, severely compressing search diversity and exacerbating search errors. In contrast, our logit manipulation framework operates globally across the unified beam of size B , allowing all beam hypotheses to explore the unconstrained syntactic space simultaneously while being steered toward required vocabulary.

Computational and Memory Complexity. From a computational perspective, logit manipulation executes in $\mathcal{O}(|\mathcal{V}|)$ time per step via direct GPU vector additions on unnormalized logits, requiring zero additional memory allocations, dynamic state-bank tracking, or CPU–GPU synchronization barriers. In contrast, DBA incurs $\mathcal{O}(B \times |\mathcal{V}|)$ trie traversal and bank rebalancing overhead at every step, creating significant CPU–GPU transfer bottlenecks and leading to the $2.1\times$ to $3.3\times$ latency penalty demonstrated in Section 5.4.

Production Deployment and Serving Framework Compatibility. Because our framework operates strictly through the standardized `LogitsProcessor` interface, it can be deployed into modern high-throughput serving architectures (such as vLLM, TensorRT-LLM, and HuggingFace Text Generation Inference) with zero modifications to the underlying autoregressive decoding loop or CUDA kernels. This plug-and-play modularity represents a major operational advantage over structural beam search modifications, which require custom generation loops that are incompatible with optimized batch-inference engines.

6. Qualitative Case Studies and Failure Taxonomy

To ground our quantitative findings in concrete linguistic behavior, Table 5 presents a qualitative comparison across decoding strategies for representative benchmark sentences, followed by an analysis of observed failure modes.

6.1 Taxonomy of Failure Modes

Through systematic error analysis across all 500 test sentences, we categorize the observed failure modes in constrained decoding into four primary phenomena.

The first category involves finite-state deadlock and trailing repetition in DBA. As illustrated in Samples 1 and 3 of Table 5, when structural DBA fails to close its internal constraint trie states, it appends verbatim constraint strings at the end of otherwise complete translations (e.g., ‘...*doktor hekim hekim.*’), directly creating the 2.768 length ratio bloat.

The second category encompasses competing scheduling in multi-constraint sequences. When a sentence contains multiple required phrases with differing natural sentence positions (e.g., *uzay istasyonu* and *yörünge*), naive hard inclusion may apply simultaneous high pressure to both tokens, causing one constraint to overwrite the other or inducing localized stuttering. Dynamic anchor scheduling mitigates this by tying pressure to sequential sentence completion progress ρ_t .

Table 5. Qualitative comparison of translation outputs across decoding strategies on representative test sentences from both translation directions.

Sample 1 (EN→TR)	Source: <i>The international space station orbits the Earth.</i>
Constraints	Required: [<i>uzay istasyonu, yörünge</i>], Forbidden: [<i>dünya</i>]
Reference	Uluslararası uzay istasyonu yerin yörüngesinde dönüyor.
Unconstrained	Uluslararası uzay istasyonu Dünya'nın yörüngesinde dönüyor.
Hard Exclusion	Uluslararası uzay istasyonu yerkürenin yörüngesinde dönüyor.
Hard Inclusion	Uluslararası uzay istasyonu yörüngede dönüyor.
Soft Reward (Escal.)	Uluslararası uzay istasyonu yerin yörüngesinde dönmektedir.
HuggingFace DBA	Uluslararası uzay istasyonu Dünya'nın yörüngesinde dönüyor. uzay istasyonu yörünge yörünge.
Sample 2 (TR→EN)	Source: <i>Dedektif önemli bir ipucu buldu.</i>
Constraints	Required: [<i>evidence</i>], Forbidden: [<i>clue</i>]
Reference	The detective found crucial evidence.
Unconstrained	The detective found an important clue.
Hard Exclusion	The detective found an important lead.
Hard Inclusion	The detective found important evidence.
Soft Reward (Escal.)	The detective found crucial evidence.
HuggingFace DBA	The detective found an important evidence.
Sample 3 (EN→TR)	Source: <i>The doctor advised the patient to exercise.</i>
Constraints	Required: [<i>hekim</i>], Forbidden: [<i>doktor</i>]
Reference	Hekim hastaya egzersiz yapmasını tavsiye etti.
Unconstrained	Doktor hastaya egzersiz yapmasını tavsiye etti.
Hard Exclusion	Tabip hastaya egzersiz yapmasını tavsiye etti.
Hard Inclusion	Hekim hastaya egzersiz yapmasını tavsiye etti.
Soft Reward (Escal.)	Hekim hastaya egzersiz yapmasını tavsiye etti.
HuggingFace DBA	Hekim hastaya egzersiz yapmasını tavsiye etti doktor hekim hekim.

The third category represents syntactic template lock-in and semantic clash. When the base-line model strongly commits to an alternative syntactic template, forcing an inclusion token with hard pressure can cause grammatical clashes with adjacent articles (e.g., generating ‘*an important evidence*’ instead of ‘*important evidence*’, as seen in DBA Sample 2). Soft reward avoids this by gently steering probability mass, allowing the language model head to adjust preceding determiners organically.

The fourth category corresponds to agglutinative mismatch and suffix stripping. In Turkish targets, when an uninflected constraint lemma is forced into an accusative or locative slot without suffix expansion, the sentence becomes ungrammatical. Our morphology-aware subword expansion enables the model to select inflected surface forms while satisfying the semantic constraint.

7. Limitations, Ethical Considerations, and Future Work

7.1 Limitations

We acknowledge several methodological limitations in this study. First, while the multi-tier escalation ladder in `soft_reward_only` achieves the highest translation quality, difficult sentences requiring Tier 2 or Tier 3 retries execute up to three decoding passes, resulting in average latencies of ~ 838 ms in EN→TR. For strict real-time applications with sub-100 ms latency budgets, single-pass Hard Inclusion remains the preferred operational mode. Second, our rule-based morphological expansion covers the most frequent nominal declensions and possessive paradigms in

Turkish, but highly complex verbal affixation (such as causatives, passives, and epistemic modalities) and irregular phonetic stem alternations are not fully enumerated. Third, corpus BLEU and ChrF are evaluated against a single human reference translation per sentence; in constrained decoding, introducing mandatory synonyms often lowers reference BLEU relative to the unconstrained baseline simply because the reference utilized an alternate synonym, even when the constrained hypothesis is fully fluent.

7.2 Ethical Considerations

This study utilizes publicly available benchmark datasets from the Tatoeba translation challenge and open-source pre-trained translation models. The research does not involve human participants, personal private data, or vulnerable populations. Constrained decoding technology provides positive safety benefits by enabling system deployers to enforce strict negative filtering on toxic, derogatory, or legally hazardous terminology in production NMT pipelines.

7.3 Future Directions

Promising avenues for future research include exploring lightweight adapter networks trained via reinforcement learning to predict optimal per-token logit deltas dynamically. Additionally, evaluating logit-level constrained decoding on other morphologically rich language families (such as Uralic, Semitic, and Bantu languages) presents an important cross-lingual benchmark. Finally, extending our dynamic anchor schedule to decoder-only large language model architectures for structured generation and instruction following represents a natural evolution of this framework.

8. Conclusion

In this paper, we presented a comprehensive, beam-search-agnostic logit manipulation framework for lexically constrained neural machine translation. By operating directly on unnormalized logit vectors through standard `LogitsProcessor` interfaces, our approach eliminates the computational overhead and structural brittleness associated with multi-bank beam allocation algorithms.

Through systematic bidirectional evaluation on English↔Turkish translation, we demonstrated that curriculum-anchored soft reward with multi-tier escalation achieves the optimal quality–satisfaction balance, attaining 99.2%/98.8% satisfaction with superior BLEU preservation and natural length ratios. Hard inclusion with dynamic anchor scheduling delivers 99.6%/99.2% satisfaction while executing 2.1–3.3× faster than the HuggingFace DBA baseline. Crucially, our analysis revealed that structural DBA suffers from severe length inflation on agglutinative targets ($2.768\times$ length in EN→TR) due to finite-state beam deadlock, whereas logit-level steering maintains clean grammatical integration. These results demonstrate that decoding-time logit manipulation represents a robust, highly scalable foundation for constrained text generation in production machine translation systems.

Declaration on the Use of Artificial Intelligence (AI) Tools

Generative AI and agentic AI coding assistants (including Google DeepMind Agentic AI Coding Assistants and Anthropic Claude) were utilized during the research and manuscript preparation. Specifically, AI tools were used to assist in writing experimental code, analyzing evaluation data and results, drafting portions of the manuscript text, and formatting LaTeX source files. Manual checks, critical evaluation, verification, and editing were performed across all AI-assisted code, data analysis, and manuscript text throughout the project, and full responsibility is taken for the integrity and content of this publication.

Ethics Statement

Ethical approval was not required for this study. All experiments were conducted using publicly available benchmark datasets (the Tatoeba parallel corpus) and open-source pre-trained machine translation models. No human participants, personal data, or sensitive information were involved.

Competing Interests

The authors declare none.

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Data Availability Statement

The bilingual evaluation corpus, constraint extraction scripts, hyperparameter optimization logs, and complete experimental codebase are openly available in the project GitHub repository at: <https://github.com/Alkem0s/lexically-constrained-decoding>.

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